

The Volume-Uniform Poincaré Walls: Machine-Checked Obstructions for Flat and Fluctuation-Sector Block-Poincaré Routes to Combes–Thomas Coercivity in Lattice Yang–Mills

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Abstract

Inside a Lean 4 formalization programme for four-dimensional $SU(N_c)$ lattice Yang–Mills, we report two machine-checked *negative* results and the machine-checked infrastructure that makes them meaningful. The positive substrate is: (i) a fixed-volume Combes–Thomas chain for self-adjoint coercive finite-range lattice operators, instantiated on the flat gauge-fixed covariance of the physical shell, with coercivity constant $c = \min(1, a)/C_P$ fed by a proved fixed-volume flat Hodge/block-Poincaré inequality; and (ii) the concrete adjoint model of $SU(n)$ — $\mathfrak{su}(n)$ with the trace inner product, $\dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1$, and the isometric transport to Euclidean coordinates — so that the abstract adjoint-model interface has a concrete nontrivial inhabitant and the flat-lane results can be instantiated with the genuine matricial adjoint model. The first wall states that, under the block normalization actually used by the formalized chain, every flat Hodge/block-Poincaré constant obeys $L^d/L^2 \leq C_P$ on the fine torus of side LN' , hence the volume-uniform Poincaré gate is *provably false* for $d \geq 3$, $N_c \geq 2$, and no positive coercivity constant survives all volumes through this route. The route consumed by the fixed-volume endpoint is therefore closed, by theorem rather than by impression. The original interface left the constant-sector quotient gate open. A transverse half-period square mode now supplies a second, strictly delimited wall. For $d \geq 3$, $N_c \geq 2$, arbitrary adjoint model, and fixed coarse side $N' = 2$, the formalization proves

$$M\langle A_M, K_0 A_M \rangle = 4\|A_M\|^2, \quad M\|Q A_M\|^2 \leq \|A_M\|^2,$$

hence $M(\langle A_M, K_0 A_M \rangle + \|Q A_M\|^2) \leq 5\|A_M\|^2$ and the volume-uniform quotient gate is false at $N' = 2$. This does not establish the corresponding statement for any other coarse side. Everything stated here is checked by Lean 4 against a pinned Mathlib, with zero `sorry`, zero project axioms, and a committed axiom-oracle transcript; a theorem–artifact map and reproduction instructions are included. The results close two routes under the current flat normalization; they establish neither a necessary condition for Yang–Mills nor any part of the continuum construction or mass-gap theorem.

1 Introduction

Constructive approaches to the four-dimensional Yang–Mills mass gap in the Balaban tradition [1, 2] route spectral information through renormalization-group covariances whose kernels must decay *uniformly in the volume*. A standard engine for such decay is the Combes–Thomas

argument [3]: a self-adjoint coercive finite-range operator has an inverse whose kernel entries decay exponentially in operator norm, at a rate governed by the coercivity constant. Whoever supplies the coercivity therefore controls the physics; and in a gauge-fixed lattice formalization the natural supplier is a Poincaré-type inequality tying the field norm to a Hodge term plus a block-averaging constraint.

This paper reports what happened when that plan was executed *formally*, in Lean 4 [6] against Mathlib [7], inside a long-running formalization programme whose earlier machine-checked artifacts include a volume-uniform Wilson-loop area law via a formalized cluster expansion [8], rooted-tree majorants for polymer expansions with holes [9], a mechanized Bessel-ratio calculus for lattice gauge expansions [10, 11, 12], an exact two-dimensional SU(2) theory [13], and the audited programme map [15].

The outcome has four layers, and we state them with the labels they deserve.

Positive (fixed volume). A complete Combes–Thomas chain for coercive finite-range operators on finite metric graphs, instantiated at the flat gauge-fixed covariance of the physical shell: for every fixed volume there exist a positive rate θ and amplitude $2/c$ with $\|(K^{-1})_{q,p}\|_{\text{op}} \leq (2/c) e^{-\theta d(q,p)}$ for all bond pairs (q, p) , where $c = \min(1, a)/C_P$ and C_P is a *proved* fixed-volume flat Hodge/block-Poincaré constant (Section 3).

Positive (non-vacuity). The abstract adjoint-model interface consumed by the chain is inhabited by the *true* matricial adjoint model of SU(n): $\mathfrak{su}(n)$ as the real inner-product space of traceless skew-Hermitian matrices under the trace form, its real dimension $n^2 - 1$, and the isometric transport to $\mathbb{R}^{n^2-1}$ carrying $\text{Ad}(g)X = gXg^\dagger$ (Section 4). The flat-lane results also admit the trivial inhabitant, since only $\text{Ad}(1)$ is consumed there; the matricial construction removes the intended-model vacuity and prepares the nontrivial-background lane.

Negative (the constant-sector wall). Under the block normalization actually used by the formalized chain — the unscaled line-integral block map, which sends constants to L times constants — *every* flat Hodge/block-Poincaré constant obeys

$$\frac{L^d}{L^2} \leq C_P$$

on the fine torus of side $N = LN'$. Consequently (Section 5): the volume-uniform Poincaré gate is *false* for $d \geq 3$, $N_c \geq 2$; the Combes–Thomas coercivity fed by this route obeys $c \leq \min(1, a) L^2/L^d \rightarrow 0$; and no single $c_0 > 0$ is dominated at every volume by the route’s coercivity. The per-volume constants *do* exist, so the obstruction kills a real route, not an empty one.

Negative (the fluctuation-sector wall at $N' = 2$). The constant-sector quotient interface (Section 6) excludes the first witness and, at the interface stage, left the volume-uniform quotient gate open. Section 7 evaluates a different witness: a transverse square-wave fluctuation mode on fine side $2M$, paired with the block map from side $2M$ to the fixed coarse side 2. Its Hodge contribution is exactly $4/M$ times its norm squared and its block contribution is at most $1/M$ times its norm squared for $d \geq 3$. The resulting $5/M$ bound contradicts a constant chosen before the scale quantifier. Thus `VolumeUniformQuotientPoincareGate` $d \geq 2$ $N_c \geq 2$ ρ is false for $d \geq 3$ and $N_c \geq 2$. No statement for another coarse side is inferred.

Honest scope

This paper proves no mass gap, no volume-uniform Combes–Thomas bound, and nothing about the thermodynamic or continuum limits. Neither gate is shown to be a necessary, equivalent, or exhaustive condition for the Yang–Mills existence-and-gap problem [5]. In particular, the second

wall is restricted literally to $N' = 2$, $d \geq 3$, $N_c \geq 2$, arbitrary adjoint model, and the current flat unscaled block operator with its present coarse norm. It gives no theorem for all coarse sizes, for a rescaled or weighted block operator, or for the interacting Wilson Hessian. The results are navigational obstructions: they close two specific flat routes. They are not a quantitative measure of progress toward the Clay problem, and no percentage-of-solution interpretation is attached to them.

2 The formal setting

All objects below are Lean definitions in the repository accompanying this paper (Section 9); we give their mathematical content and their Lean names.

Cochains. Fix $d, N, N_c \in \mathbb{N}$, all positive. Sites form the discrete torus $\Lambda_N = (\mathbb{Z}/N)^d$ (`FinBox d N`); positive bonds are pairs $b = (x, k)$ of a site and a direction (`PhysicalBond d N`). A *physical one-cochain* is a map A from positive bonds to the internal coordinate space $\mathbb{R}^{N_c-1}$ (`SUNLieCoord Nc`), with the ℓ^2 inner product in both the bond and internal indices (`PhysicalGaugeOneCochain d N Nc`).

The block map. For $N = LN'$, the flat block constraint Q : fine cochains (side LN') $\rightarrow$ coarse cochains (side N') assigns to a coarse bond the *unscaled* line integral (sum) of the L fine bond values along the corresponding block line (`flatBlockConstraintQCLM`, built on `linAvg`). The normalization fact that drives everything below is exact (`flatBlockConstraintQCLM_constant`):

$$Q(A_v) = L \cdot A_v^{\text{coarse}}, \quad (1)$$

where A_v denotes the direction-wise constant cochain $A_v(x, k) = v_k$ for a fixed family $v: \{1, \dots, d\} \rightarrow \mathbb{R}^{N_c-1}$ (`constantPhysicalGaugeOneCochain`).

The flat Hodge operator. At the trivial gauge background, $K_0 = D_1^\dagger D_1 + D_0 D_0^\dagger$ (`flatGaugeHodgeK0CLM`), so that $\langle A, K_0 A \rangle = \|\text{curl } A\|^2 + \|\text{div } A\|^2$ with the standard lattice curl on plaquettes and backward divergence on sites (`flatGaugeHodgeK0_inner_right`). Constants are flat-harmonic: $K_0 A_v = 0$ (`flatGaugeHodgeK0CLM_constantPhysicalGaugeOneCochain`).

The Poincaré predicate. For an adjoint model ρ (Section 4) and $C_P \in \mathbb{R}$, `FlatGaugeHodgePoincare d L N' Nc ρ CP` is the statement

$$0 < C_P \quad \wedge \quad \forall A: \|A\|^2 \leq C_P (\langle A, K_0 A \rangle + \|QA\|^2), \quad (2)$$

over fine cochains at side LN' . The repository proves, at every fixed volume, that such constants exist (`exists_flatGaugeHodgePoincare`); this non-vacuity is what makes the negative results of Section 5 statements about a live route.

3 The fixed-volume Combes–Thomas endpoint

The Combes–Thomas chain is formalized for abstract self-adjoint coercive finite-range operators on finite metric graphs: the tilt algebra and tilted entry bounds, the block Schur bound,

coercivity survival under tilting at the budget $M(e^{\theta R} - 1)N_R \leq c/2$, the tilted inverse with $\|K_\theta^{-1}\| \leq 2/c$, and the kernel extraction at a root (`PhysicalCoerciveCombesThomas*.lean`). Instantiated at the flat gauge-fixed covariance of the physical shell (stencils and locality proved, not assumed), the endpoint reads:

Theorem 3.1 (fixed-volume Combes–Thomas; `flatGaugeFixedCovariance_CT_fixedVolume`, `exists_flatGaugeFixedCovariance_CT_fixedVolume`). *Fix the volume parameters (d, L, N') , a mass $a > 0$, and a flat Hodge/block-Poincaré constant C_P as in (2). Let $c = \min(1, a)/C_P$. Then there exists $\theta > 0$ meeting the explicit tilt budget such that for all bond pairs (q, p) and every internal vector v ,*

$$\|(K^{-1})_{q,p} v\| \leq \frac{2}{c} e^{-\theta d(q,p)} \|v\|,$$

equivalently $\|(K^{-1})_{q,p}\|_{\text{op}} \leq (2/c) e^{-\theta d(q,p)}$ — the form quantified literally by `PhysicalCovarianceExponentialKernelBound`.

Fixed volume means fixed volume: C_P , hence c and θ , may depend on (d, L, N', a) . The question the rest of this paper answers is whether *this* c can be made volume-uniform. It cannot.

4 The true adjoint model of $\text{SU}(n)$

The chain consumes an abstract interface `SUNAdjointModel n`: a family of isometries $\text{Ad}(g)$ of $\mathbb{R}^{n^2-1}$ indexed by $g \in \text{SU}(n)$ with $\text{Ad}(1) = \text{id}$, $\text{Ad}(gh) = \text{Ad}(g)\text{Ad}(h)$, and inner-product invariance. A trivial inhabitant ($\text{Ad}(g) = \text{id}$) exists and suffices for the flat lane, where only $\text{Ad}(1)$ is consumed; leaving matters there would invite a vacuity objection the moment a non-trivial background appears. The repository therefore constructs the genuine object:

Theorem 4.1 (the matricial adjoint model; `matrixSUNAdjointModel`). *Let $\mathfrak{su}(n)$ be the real vector space of traceless skew-Hermitian $n \times n$ complex matrices with the trace form $\langle X, Y \rangle = \text{Retr}(X^\dagger Y)$ (`SUNAdjointMatrixSubstrate`, `SUNAdjointInnerSpace`). Then $\dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1$ (`finrank_suLie`), the reindexed orthonormal-basis representation gives a linear isometric equivalence $\mathfrak{su}(n) \simeq \mathbb{R}^{n^2-1}$ (`suLieCoordIso`), and conjugating the adjoint action $X \mapsto gXg^\dagger$ by this isometry yields a `SUNAdjointModel n` whose action laws and inner-product invariance are proved, not hypothesized.*

The dimension count is itself formalized (Hermitian/skew decomposition, multiplication-by- i equivalence, ambient real dimension $2n^2$, and rank-nullity through the imaginary-trace functional with explicit surjectivity witness $i \cdot \mathbf{1}$). The coordinate choice is the non-canonical standard orthonormal basis; any other choice conjugates the model by a fixed orthogonal map, and nothing downstream depends on it.

5 The wall

Everything in this section is proved in `PhysicalGaugeFlatPoincare.lean` (the fixed-volume audit) and `PhysicalPoincareWall.lean` (the volume-uniform obstruction).

5.1 The constant sector forces the constant

Lemma 5.1 (exact sector normalization; `flatBlockConstraint_controls_constantSector`). *For every v ,*

$$\|A_v\|^2 = \frac{L^d}{L^2} \|QA_v\|^2.$$

Proof. $\|A_v\|^2 = N^d \sum_k \|v_k\|^2$ with $N = LN'$ (`norm_sq_constantPhysicalGaugeOneCochain`), while by (1), $\|QA_v\|^2 = L^2 N'^d \sum_k \|v_k\|^2$ (`flatBlockConstraintQCLM_constant_norm_sq`); divide. $\square$

Theorem 5.2 (the wall, fixed volume; `flatPoincare_constantSector_lower_bound`). *Let $N_c \geq 2$ and let C_P satisfy (2). Then*

$$\frac{L^d}{L^2} \leq C_P.$$

Proof. For $N_c \geq 2$ the internal space $\mathbb{R}^{N_c^2-1}$ is nontrivial, so there is v with $\sum_k \|v_k\|^2 > 0$ (`exists_pos_constantSector`; for $N_c = 1$, $\mathfrak{su}(1) = 0$ and the sector is empty — the hypothesis is stated, not hidden). Apply (2) to A_v : the Hodge term vanishes ($K_0 A_v = 0$), so $\|A_v\|^2 \leq C_P \|QA_v\|^2$, and Lemma 5.1 gives the claim after cancelling $\|QA_v\|^2 > 0$. $\square$

Corollary 5.3 (linear form; `flatPoincare_linear_lower_bound`). *If moreover $d \geq 3$, then $L \leq C_P$. (For $d = 4$ the forced growth is L^2 ; the linear form is all the divergence the obstruction needs. $d = 2$ is the scale-invariant exemption $L^d/L^2 = 1$ and is deliberately not claimed.)*

5.2 The route's coercivity dies with the volume

Theorem 5.4 (coercivity upper bound; `flatPoincare_coercivity_upper_bound`). *For $a > 0$, $N_c \geq 2$, and any C_P as in (2),*

$$\frac{\min(1, a)}{C_P} \leq \min(1, a) \frac{L^2}{L^d}.$$

Theorem 5.5 (no volume-uniform coercivity through flat Poincaré). *Lean:*

`no_volumeUniform_coercivity_via_flatPoincare`.

Let $d \geq 3$, $N_c \geq 2$, $a > 0$. Suppose $c_0 \in \mathbb{R}$ is dominated at every volume: for every $L \in \mathbb{N}$ there exists $C_P(L)$ satisfying (2) at side $(L+1)N'$ with $c_0 \leq \min(1, a)/C_P(L)$. Then $c_0 \leq 0$.

Proof. If $c_0 > 0$, choose (Archimedes) n with $n > \min(1, a)/c_0$. By Corollary 5.3, $C_P(n) \geq n + 1 > \min(1, a)/c_0$, whence $\min(1, a)/C_P(n) < c_0$, contradicting domination at volume $n + 1$. $\square$

Theorem 5.6 (the gate is false; `volumeUniformFlatPoincareGate_false`). *For $d \geq 3$, $N_c \geq 2$, there is no single $C_P > 0$ satisfying (2) at every volume $L + 1$: the proposition `VolumeUniformFlatPoincareGate` — with the constant quantified before the volume — is provably false.*

Remark 5.7 (non-vacuity of the killed route; `perVolume_flatPoincare_family`). *The hypothesis family of Theorem 5.5 is inhabited: per-volume constants exist at every volume. The obstruction closes a route that is genuinely open at each fixed volume.*

Remark 5.8 (quantifier discipline). *The gate places its constant before the volume quantifier. The reversed order — for every volume there exists a constant — is exactly the (true) per-volume statement, and conflating the two is the standard way such obstructions get lost in prose. The formalization makes the distinction type-level.*

6 The constant-sector quotient interface

The wall witness lives in one sector. The natural continuation restricts the Poincaré question to its orthogonal complement — the physical fluctuation space. The repository formalizes the *interface* (`PhysicalPoincareSectorQuotient.lean`); we use the word “interface” deliberately: no quotient type or packaged orthogonal-complement submodule is constructed, and the predicate is a restriction to fluctuation cochains.

Definition 6.1 (fluctuation cochains; `IsFluctuationCochain`). A is a fluctuation cochain if $\langle A_v, A \rangle = 0$ for every constant generator A_v — equivalently, A is orthogonal to the span of the constant sector. (The generator-wise form is chosen for formal robustness.)

Proposition 6.2 (interface facts). (i) *The constant sector is flat-harmonic* (`constantSectorLin_harmonic`) — which is why only the block term sees it, with the normalization of Lemma 5.1. (ii) *The full predicate (2) implies its restriction to fluctuation cochains at the same constant* (`quotientFlatPoincare_of_flatPoincare`); the restricted statement is strictly weaker, so Theorem 5.6 does not refute the quotient gate. (iii) *Restricted constants exist at every fixed volume* (`exists_quotientFlatPoincare`). (iv) (Non-transfer.) *A nonzero constant cochain is not a fluctuation cochain* (`constant_not_fluctuation`): the wall witness is excluded from the fluctuation space by construction.

Remark 6.3 (what the interface stage established). *At the interface stage the volume-uniform quotient gate* (`VolumeUniformQuotientPoincareGate`) *was neither proved nor refuted. Non-transfer removed the $W-1$ witness, not the possibility of a second wall. The present revision preserves that historical statement and adds a different fluctuation witness. Its endpoint is conclusive only at fixed coarse side $N' = 2$.*

7 The low-mode falsifier and the second wall at $N' = 2$

A *one-sided falsifier* for the quotient gate needs an explicit fluctuation family on which the quadratic form of (2) can be evaluated exactly. Four modules now implement the complete chain: construction, Hodge evaluation, block response, and endpoint.

Definition 7.1 (the half-period (two-interface) square-wave mode; `squareModeCochain`). On even side lengths $N = 2M$ ($M \geq 1$), with oscillation direction j , bond direction i , and internal vector w :

$$A(x, k) = \begin{cases} \varepsilon(x_j) w, & k = i, \\ 0, & k \neq i, \end{cases} \quad \varepsilon(m) = \begin{cases} +1, & m < M, \\ -1, & m \geq M, \end{cases}$$

(`squareSign`). The square profile is chosen over the cosine so that every identity below is an exact finite ± 1 computation; odd side lengths are out of scope by declared parameterization (a falsifier needs only *some* volume sequence).

Proposition 7.2 (exact identities). (i) $\sum_m \varepsilon(m) = 0$ (`sum_squareSign`), hence A is a fluctuation cochain (`squareModeCochain_isFluctuation`) — exact orthogonality, no approximation. (ii) $\|A\|^2 = (2M)^d \|w\|^2$ exactly (`norm_sq_squareModeCochain`). (iii) For $w \neq 0$ the mode equals no constant cochain (`squareModeCochain_ne_constant`), and the fluctuation space is nontrivial for $N_c \geq 2$ (`exists_nonzero_fluctuation`).

For the endpoint choose distinct directions $i \neq j$, let $A_M = \text{squareModeFine } d \ M \ N_c \ i \ j \ w$, and apply the block map with block length M and coarse side 2. The fine side is therefore $M \cdot 2 = 2M$. The Hodge calculation gives

$$M \langle A_M, K_0 A_M \rangle = 4 \|A_M\|^2. \quad (3)$$

The block and fine norms are exact:

$$\|QA_M\|^2 = 2^d M^2 \|w\|^2, \quad \|A_M\|^2 = (2M)^d \|w\|^2.$$

Thus, when $d \geq 3$ and $M \geq 1$,

$$M \|QA_M\|^2 \leq \|A_M\|^2. \quad (4)$$

Combining (3) and (4) without division produces

$$M (\langle A_M, K_0 A_M \rangle + \|QA_M\|^2) \leq 5 \|A_M\|^2. \quad (5)$$

Dependency record for the four capstone theorems

Lean theorem	Mathematical role and immediate dependencies
<code>squareMode_mul_hodge_eq_four_norm_sq</code>	Statement: (3). Inputs: <code>flatGaugeHodgeK0_squareMode_energy_mul_side</code> and the definitional identification of <code>squareModeFine</code> with the square-mode cochain. Role: exact Hodge contribution.
<code>squareMode_mul_block_le_norm_sq</code>	Statement: (4). Hypotheses: $d \geq 3, M \geq 1, i \neq j$. Inputs: <code>norm_sq_flatBlockConstraintQCLM_squareModeFine_of_ne</code> and <code>norm_sq_squareModeFine</code> . Role: transverse block control.
<code>squareMode_mul_totalEnergy_le_five_norm_sq</code>	Statement: (5). Inputs: the preceding two theorems. Role: division-free Rayleigh-numerator bound.
<code>volumeUniformQuotientPoincareGate_two_false</code>	Statement: the quotient gate is false at coarse side 2. Hypotheses: $d \geq 3, N_c \geq 2; \rho$ arbitrary. Inputs: (5), fluctuation membership, exact norm positivity, a nonzero internal vector, and the Archimedean choice of M . Role: quantifier-respecting endpoint.

Theorem 7.3 (the quotient gate is false at coarse side two). Lean:

`volumeUniformQuotientPoincareGate_two_false`.

For $d \geq 3, N_c \geq 2$, and every adjoint model ρ ,

$$\neg \text{Gate}_{\text{quot}}(d, 2, N_c, \rho).$$

Proof. Assume a gate witness $C_P > 0$ whose scale quantifier comes after the choice of C_P . Choose $M \in \mathbb{N}$ with $M > 5C_P$, choose distinct directions and a nonzero internal vector, and instantiate the gate at the corresponding fine side $2M$. The mode is a fluctuation and has positive norm. Multiplying the Poincaré inequality by M and using (5) gives

$$M \|A_M\|^2 \leq 5C_P \|A_M\|^2.$$

Positivity permits cancellation, so $M \leq 5C_P$, a contradiction. The formal proof performs this argument without dividing by $\|A_M\|^2$. $\square$

Remark 7.4 (scope and one-sided discipline). *This is a second formally certified wall for the current flat quotient-Poincaré route, at fixed coarse side $N' = 2$. It does not establish falsity for another coarse side, necessity or exhaustiveness of the gate for Yang–Mills, or any statement about the mass gap. The falsifier has fired in its conclusive direction; had its quotient stayed bounded, that fact alone would not have proved the gate.*

8 Theorem–artifact map

Statement	Lean artifact (module under YangMills/RG/)
CT endpoint, fixed volume	<code>flatGaugeFixedCovariance_CT_fixedVolume</code> (PhysicalShellCombesThomasEndpoint)
CT positive-rate witness	<code>exists_flatGaugeFixedCovariance_CT_fixedVolume</code> (PhysicalShellCombesThomasEndpoint)
$\dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1$	<code>finrank_suLie</code> (SUNAdjointDimension)
Isometric transport	<code>suLieCoordIso</code> (SUNAdjointModelInstance)
True adjoint model	<code>matrixSUNAdjointModel</code> (SUNAdjointModelInstance)
Sector witness ($N_c \geq 2$)	<code>exists_pos_constantSector</code> (PhysicalPoincareWall)
Wall, fixed volume	<code>flatPoincare_constantSector_lower_bound</code> (PhysicalPoincareWall)
Wall, linear form ($d \geq 3$)	<code>flatPoincare_linear_lower_bound</code> (PhysicalPoincareWall)
Coercivity upper bound	<code>flatPoincare_coercivity_upper_bound</code> (PhysicalPoincareWall)
No volume-uniform c_0	<code>no_volumeUniform_coercivity_via_flatPoincare</code> (PhysicalPoincareWall)
Full gate false	<code>volumeUniformFlatPoincareGate_false</code> (PhysicalPoincareWall)
Per-volume non-vacuity	<code>perVolume_flatPoincare_family</code> (PhysicalPoincareWall)
Fluctuation predicate	<code>IsFluctuationCochain</code> (PhysicalPoincareSectorQuotient)
Full $\Rightarrow$ quotient	<code>quotientFlatPoincare_of_flatPoincare</code> (PhysicalPoincareSectorQuotient)
Non-transfer of first witness	<code>constant_not_fluctuation</code> (PhysicalPoincareSectorQuotient)
Quotient-gate predicate	<code>VolumeUniformQuotientPoincareGate</code> (PhysicalPoincareSectorQuotient)
Square profile, zero sum	<code>sum_squareSign</code> (PhysicalPoincareLowModeFalsifier)
Mode is fluctuation	<code>squareModeCochain_isFluctuation</code> (PhysicalPoincareLowModeFalsifier)
Mode norm, exact	<code>norm_sq_squareModeCochain</code> (PhysicalPoincareLowModeFalsifier)
Hodge identity	<code>squareMode_mul_hodge_eq_four_norm_sq</code> (PhysicalPoincareLowModeEndpoint)
Block inequality	<code>squareMode_mul_block_le_norm_sq</code> (PhysicalPoincareLowModeEndpoint)
Total-energy inequality	<code>squareMode_mul_totalEnergy_le_five_norm_sq</code> (PhysicalPoincareLowModeEndpoint)
Quotient gate false at $N' = 2$	<code>volumeUniformQuotientPoincareGate_two_false</code> (PhysicalPoincareLowModeEndpoint)

Every listed name is exercised by the axiom-oracle script. In particular, all sixteen W-3b/W-3c/endpoint queries, including the four capstone rows, print exactly [propext, Classical.choice, Quot.sound].

9 Reproducibility

Repository. <https://github.com/lluiseriksson/THE-ERIKSSON-PROGRAMME> (review branch `poincare-wall-w3-review`). The first artifact remains frozen at source checkpoint `12ca1a87`, sealed by `64969b35`. The W-3 review is reconstructed from that seal. Its isolated Lean and oracle checkpoint is `90ed3585`; no result or document from a different paper lane is included in the review delta. The scientific manuscript checkpoint is `009da5b0`.

Toolchain. Lean `leanprover/lean4:v4.29.0-rc6`; Mathlib pinned to commit `0764272048...` (lakefile and manifest agree). Build: `lake build YangMillsCore` — expected Build completed successfully (8413 jobs). The increase of three jobs is exact: the falsifier module already belonged to the frozen tree, while the Hodge, block, and endpoint modules are new.

Axiom oracle. `lake env lean oracle_check.lean` executes 2210 `#print axioms` invocations: the frozen 2194 plus sixteen individually identified W-3 queries. A separate focal run of those sixteen returns exactly `[propext, Classical.choice, Quot.sound]` for every target, with zero `sorryAx` and no project axiom. The full transcript is `docs/oracle-transcripts/ORACLE-20260714-90ed3585.txt`: 2124 results use the full trio, 49 use `[propext, Quot.sound]`, 17 use `[propext]`, and 20 are axiom-free. The LF-normalized raw output has SHA-256 `09044831B051646510F828C6E1C3A45B9BA3503482DB65866C695259E43DD75C`.

Continuation from this submission

The Hodge evaluation, block estimate, division-free total-energy bound, and endpoint at $N' = 2$ are now theorems, not future work. The surviving questions are deliberately separate: other coarse sides; rescaled or weighted block operators with their normalization re-audited from first principles; and coercivity from the interacting Hessian. No claim that any of these routes succeeds is made.

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